Effects of Temperature on Growth and DNA Methylation Patterns in Spirodela polyrhiza

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Abstract

As global temperatures continue to rise due to climate change, understanding how environmental thermal stress affects the growth, reproduction, and survival of aquatic plant species becomes increasingly critical. This study measured the effect of temperature on the freshwater species *Spirodela polyrhiza* to determine whether it produced any significant changes in growth and DNA methylation patterns, predicting that elevated temperatures would inhibit growth and alter methylation at heat-responsive loci. Eighteen microplate replicates containing individual turions in Hoagland solution were incubated at 20°C, 30°C, and 40°C for 21 days, with frond count and percent surface coverage measured weekly via ImageJ. DNA was extracted, bisulfite converted, and analyzed for CpG methylation at the HSFA1 and ONSEN loci using Methylation-Specific PCR. While growth increased across all groups over time, higher temperatures trended toward reduced frond count and coverage without reaching statistical significance. In contrast, CpG methylation at both loci decreased significantly with increasing temperature, with all pairwise comparisons reaching significance. These findings suggest that thermal stress drives epigenetic reorganization in *S. polyrhiza* before physical growth effects become statistically detectable, highlighting the potential for heritable methylation changes to broadly affect the molecular regulation of aquatic plant species and the freshwater ecosystems they support.

Introduction

Climate change has been one of the leading driving factors behind the increase in global temperatures over the past century. This phenomenon has been linked to an increase in extreme weather conditions, population declines, and poses a severe threat to different species throughout the globe. Among the various natural and human-made factors that contribute to climate change, some of the largest contributors are those that deplete the ozone layer in the atmosphere, such as fossil fuel combustion, pollution, and intensive agriculture. These environmental changes are particularly significant for plant species, which are highly sensitive to rapid changes in temperature. Collectively, these issues are exposing aquatic environments to extreme temperatures, ocean acidification, and microplastic pollution (Van Antro et al., 2023). To survive these shifts, many plants rely on adaptive mechanisms such as epigenetic regulation, including DNA methylation, which allows them to modify gene expression in response to temperature-driven environmental stress.

DNA methylation is one of the various genetic regulatory adaptations that help species survive extreme environmental conditions. Methylation is a mechanism in which a methyl group is added to cytosines to aid in suppressing transposable element activity and regulating specific genes (Van Antro et al., 2023). DNA methylation patterns can be heritable, meaning that these regulatory changes may persist across

generations. Not only may environmental factors affect the DNA methylation profiles of an individual organism, but they may also impact offspring, leading to a “memorized” epigenetic response to environmental stress (Van Antro et al., 2023).

Spirodela polyrhiza is a species of duckweed characterized by rapid growth, relatively large frond size, and unique turion-forming properties. Turions are smaller structures with thick cell walls and high storage capacity, allowing them to sink to the bottom of bodies of water (Appenroth & Nickel, 2010). The formation of these turions is influenced by environmental factors, which are divided into two classes: those that influence growth rate, such as sugars and photosynthesis, and stress factors that inhibit growth, which are of particular interest in this study. This duckweed species stands out due to its rapid growth rate, existing studies on DNA methylation patterns in Spirodela and related species, and the critical role of turions in survival under extreme conditions. These characteristics make S. polyrhiza an ideal model organism for studying temperature-induced epigenetic responses, especially in the context of increasing global temperatures due to climate change.

While extensive research exists on DNA methylation and climate change, limited work has focused on species-specific mechanisms, particularly in aquatic plants and their winter buds, such as turions. Lopez et al. (2024) analyzed heat stress responses in Fragaria vesca (strawberry) using controlled growth conditions, DNA methylation, and genomic transcriptomic profiling. This study presents data and insights that may be useful in understanding how plants biologically adapt to rising temperatures, revealing optimal growth conditions and evolutionary adaptations. Understanding how aquatic plants like Spirodela polyrhiza epigenetically respond to temperature stress have implications for both agricultural resilience and long term environmental monitoring (Kakoulidou et al., 2021). Additionally, linking temperature stress to parental methylation patterns may be beneficial for assessing the long term impacts of global warming on ecosystems through plant epigenetic profiles(Van Antro et al., 2023).

This experiment aims to further investigate previously established claims about inhibiting growth factors in S. polyrhiza and their impact on turion formation and DNA methylation patterns (Kuehdorf & Appenroth, 2011). Growth and stress responses were assessed across temperatures ranging from 20°C to 40°C. Plant performance was measured through the number of fronds present and the surface area of the duckweed. Furthermore, disruptions from environmental stressors have been shown to be difficult to detect without molecular level analysis, emphasizing the value of DNA methylation as an indicator of growth in S. polyrhiza (Dainelli et al., 2024). The Null hypothesis (Ho) is that Temperature will not have a significant effect on turion germination success or DNA methylation occurrence. The Alternative hypothesis (Ha) is that Temperature will have a significant effect on turion germination success and DNA methylation occurrence. It is predicted that higher temperatures will inhibit frond growth and turion germination while altering DNA methylation patterns in S. polyrhiza. This prediction is based on prior evidence that high temperatures activate epigenetic regulatory mechanisms as a survival response in aquatic plants (Kuehdorf & Appenroth, 2011). The results of this study may support future research on duckweeds and related aquatic plant species by providing data on stress mechanisms, as well as insights into how climate change and environmental pollution influence micro-evolutionary processes and epigenetic regulation in aquatic ecosystems.

Methodology

Experimental Design:

Three treatment groups of six replicates were placed into 18 microplates in a sterile environment. These groups were placed into 20°C, 30°C, and 40°C incubation chambers, and data on the physical growth were collected every week. Each microplate was filled with 10 mL of Hoagland solution (HS) as the choice of the growth medium for the experiment due to its high reliability in plant growth of a large variety of plant species, including S. polyrhiza (Hoagland & Arnon, 1950). A single turion was placed into each replicate using a heated inoculation loop.

Data Collection:

Top-down snapshots of the replicates were taken and imported into the software tool ImageJ every week for the next 3 weeks. The multi-point tool was used to count the number of fronds formed; the freehand tool measured frond surface area with the line tool set to a scale of 34.8 mm, and the percentage of the microplate area occupied by the duckweeds was calculated. These measurements of growth are important as they often serve as indicators of turion development and physiological stress responses.

Genetic Sequencing:

Following the 21-day growth period, high-quality turion DNA was extracted and isolated using the Quick-DNA Plant/Seed Miniprep Kit. Extracted DNA was then stored for 7 days before undergoing bisulfite conversion with the EZ DNA Methylation Kit. During the 7 day gap, the DNA samples were stored under optimal conditions to prevent degradation from affecting the results. Bisulfite conversion provided a way to properly quantify methylation by converting unmethylated cytosines to uracils, leaving methylated cytosines unchanged and thus distinguishable during the sequencing analysis. Methylation status at the ONSEN and HSFA1 loci was then assessed using Methylation-Specific PCR (MSP), which detects the presence or absence of methylation at target CpG sites.

Data Analysis:

The frond count and percent coverage as well as the methylation site count were the variables used to indicate growth. Line graphs were generated to visualize growth over time, and bar graphs of treatment group means were created for methylation comparisons. All figures and tables were generated using Python, through the Matplotlib library, within Google Colab. The statistical test used was a two-way Analysis of Variance (ANOVA) run in Google Colab using a Jupyter notebook. Following the ANOVA, a Tukey Honestly Significant Difference (HSD) post-hoc test was applied to identify which specific treatment pairs had a significant difference, and the ANOVA indicates that a difference exists somewhere within the groups with a threshold of $p_value < 0.05$ to serve as a significance threshold. For methylation, a one-way ANOVA followed by a Tukey HSD post-hoc test were used to compare CpG methylation levels across the three temperature treatments.

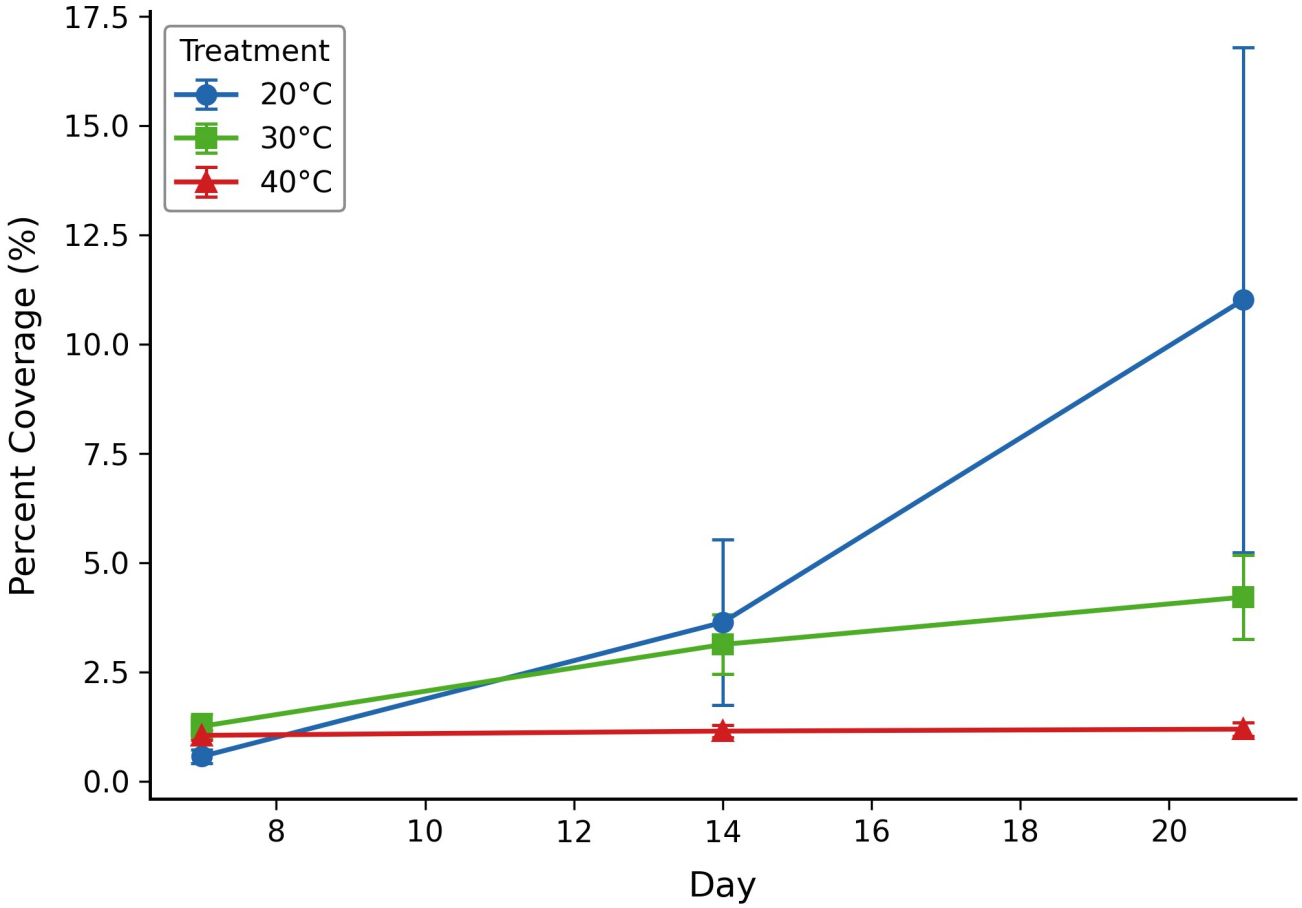
Results

The percent coverage and number of fronds of duckweed increased over the course of the experiment across all temperature treatments (Figures 1 and 2). Growth differed by temperature, with the 20°C group consistently showing the highest values for both variables, followed by the 30°C group, and the 40°C group showing the least growth overall. A two-way ANOVA revealed that day was the only statistically significant factor for both percent coverage and number of fronds ($F = 3.655$, $p = 0.0338$), while temperature treatment and the temperature-by-day interaction were not significant. Post-hoc Tukey HSD analysis confirmed no significant pairwise differences between any temperature treatments for either variable, with all adjusted-p values

exceeding 0.05 (Tables 1 and 2). These results suggest that while duckweed growth increased over time, higher temperatures trended toward reduced growth.

Figure 1

Percent Coverage of Duckweed Across Temperature Treatments Over Time.



Percent coverage represents the proportion of the well surface area occupied by duckweed. Measurements were recorded every 7 days over a 21-day period (days 7, 14, and 21). Error bars indicate standard error.

Table 1

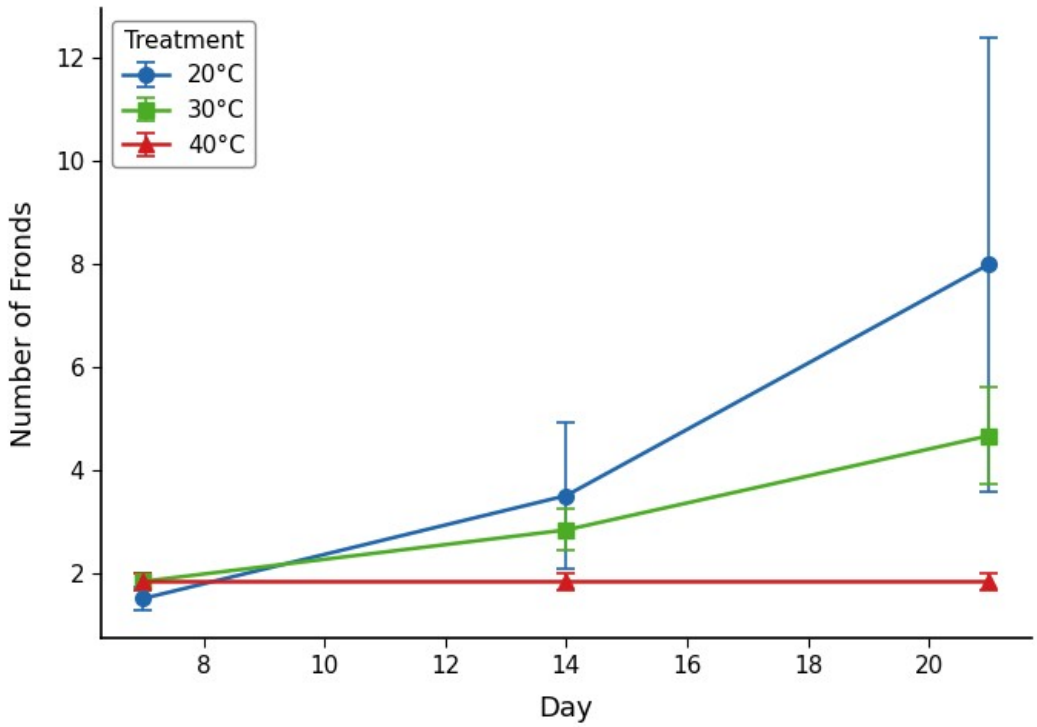
Tukey HSD: Percent Coverage by Treatment

Percent coverage represents the percentage of the well surface area occupied by duckweed. Group1 and group2 represent the two groups being compared, meandiff is the difference between their means, p-adj is the p-value adjusted for multiple comparisons using Tukey's Honestly Significant Difference (HSD) method (FWER = 0.05), and reject indicates whether the difference is statistically significant (True: $p < 0.05$; False: not significant). All comparisons resulted in non-significant differences.

Group 1	Group 2	Mean Diff.	p-adj	Lower	Upper	Reject H_0
20°C	30°C	-2.2065	0.4529	-6.6095	2.1965	False
20°C	40°C	-3.9436	0.0876	-8.3466	0.4594	False
30°C	40°C	-1.737	0.61	-6.14	2.666	False

Figure 2

Number of Fronds Across Temperature Treatments Over Time.



Frond count represents the total number of individual duckweed fronds present in each well. Measurements were recorded every 7 days over a 21-day period (days 7, 14, and 21). Error bars indicate standard error.

Table 2

Tukey HSD: Number of Fronds by Treatment.

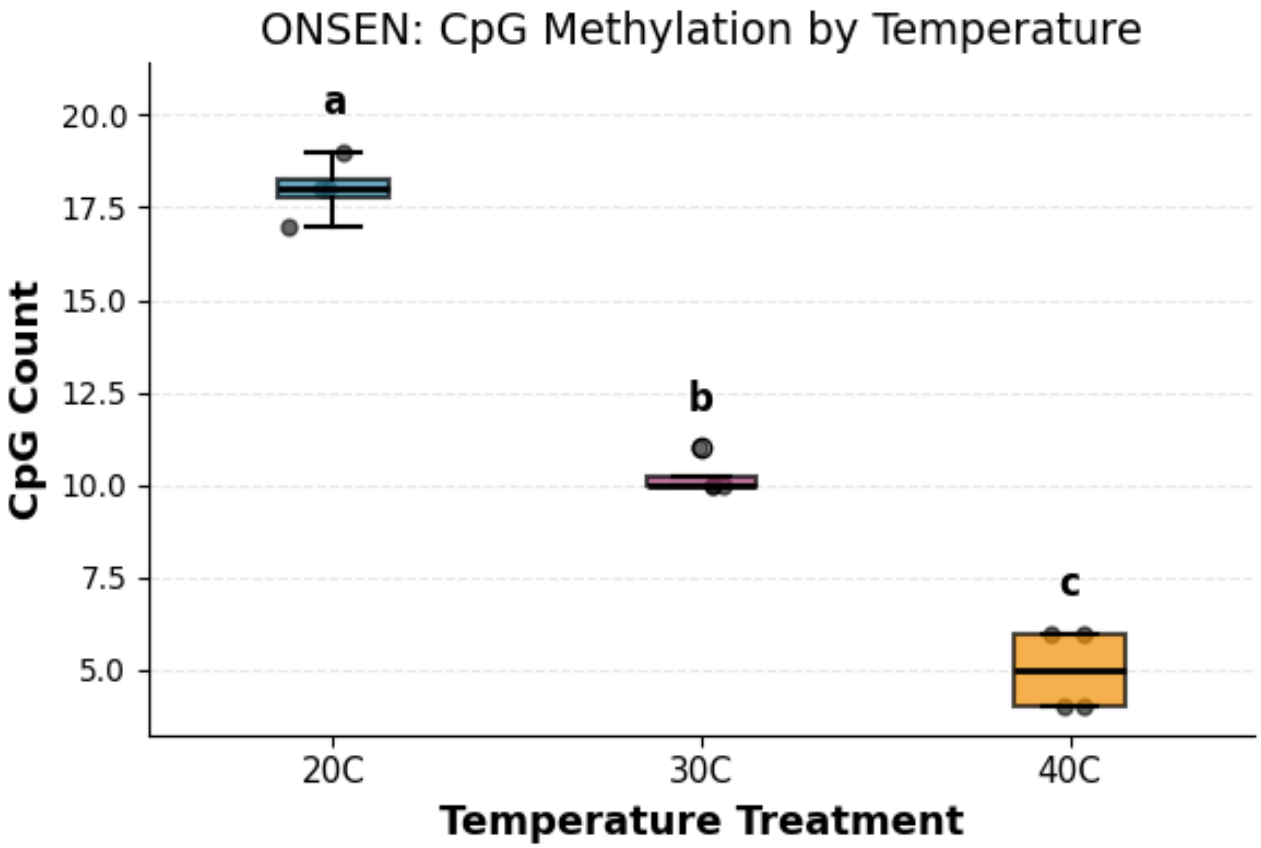
Number of fronds represents the total frond count per well for each temperature treatment. Group1 and group2 represent the two groups being compared, meandiff is the difference between their means, p-adj is the p-value adjusted for multiple comparisons using Tukey's Honestly Significant Difference (HSD) method (FWER = 0.05), and reject indicates whether the difference is statistically significant (True: $p < 0.05$; False: not significant). All comparisons resulted in non-significant differences.

Group 1	Group 2	Mean Diff.	p-adj	Lower	Upper	Reject H_0
20°C	30°C	−1.2222	0.6407	−4.4884	2.0439	False
°C	40°C	−2.5000	0.1646	−5.7662	0.7662	False
30°C	40°C	−1.2778	0.6150	−4.5439	1.9884	False

CpG methylation at the ONSEN gene varied across temperature treatments, with methylation levels decreasing as temperature increased (Fig. 3). The 20°C group showed the highest number of methylated CpG sites, followed by the 30°C group, and the 40°C group showing the least methylation overall. A post-hoc Tukey HSD analysis confirmed that all pairwise comparisons were statistically significant (Table 3). Specifically, the 20°C group differed significantly from both the 30°C group (meandiff = -2.5, $p = 0.0016$) and the 40°C group (meandiff = -4.0, $p < 0.001$), and both the 30°C and 40°C groups differed significantly from each other (meandiff = -1.5, $p = 0.0316$). This pattern suggests that higher temperature treatments are associated with reduced DNA methylation at the ONSEN and HSFA1 gene.

Figure 3

CpG Methylation by Temperature (ONSEN)



CpG methylation represents the number of methylated CpG sites under three temperature treatments (20°C, 30°C, 40°C). Boxplots show the median, interquartile range, and individual data points. Groups with different letters are significantly different. All comparisons were significant ($p < 0.05$).

Table 3

Tukey HSD: CpG Methylation by Treatment (HSFA1).

CpG methylation represents the number of methylated CpG sites for the HSFA1 gene under each temperature treatment. Group1 and group2 represent the two groups being compared, meandiff is the difference between their means, p-adj is the p-value adjusted for multiple comparisons using Tukey's Honestly Significant Difference (HSD) method (FWER = 0.05), and reject indicates whether the difference is statistically significant (True: $p < 0.05$; False: not significant). All comparisons in this dataset were significant.

Group 1	Group 2	Mean Diff.	p-adj	Lower	Upper	Reject H_0
20°C	30°C	−2.5	0.0016	−3.8567	-1.1433	True
20°C	40°C	−4.0	0.0	-5.3567	−2.6433	True
30°C	40°C	−1.5	0.0316	−2.8567	−0.1433	True

Discussion

In this study, we hypothesized that temperature would have a significant effect on turion germination success and DNA methylation occurrence in Spirodela polyrhiza, predicting that plants exposed to elevated temperatures would exhibit reduced growth. Our results indicated that while higher temperatures trended toward reduced frond count and percent coverage, these differences did not reach statistical significance, and the null hypothesis for growth could not be rejected. However, CpG methylation at the ONSEN loci decreased significantly as the temperature increased, leading to rejection of the null hypothesis for DNA methylation. These findings reveal that temperature is not an insignificant variable in duckweed development, but rather a meaningful environmental stressor capable of influencing epigenetic regulation.

The reduction in frond number observed in the heat stressed groups of Spirodela polyrhiza likely reflects thermal inhibition of essential cellular processes, including enzyme activity, photosynthetic output, and cell division, as elevated temperatures push metabolic function beyond optimal operating ranges in aquatic macrophytes. This prediction lines up with evidence that exposure to high temperatures can alter both DNA methylation patterns and growth-related phenotypes in clonal aquatic plants, disrupting the molecular regulation that normally

coordinates development (Naydenov et al., 2015). Similarly, the reduced methylation detected at stress responsive loci such as ONSEN in heat stressed groups suggests that temperature stress triggers a reorganization of the epigenome through the activation of key components of the DNA methylation pathway, such as DRM2 and NRPD1, and DNA demethylation via the activation of demethylase enzymes (Naydenov et al., 2015). Therefore, the plant's methylation response may function as a form of molecular stress signaling as an adaptive mechanism to sudden environmental change.

One unexpected finding was that standard error for the 40°C treatment group on day 21 for the turion count and percent coverage (Figures 1 and 2) was compromised. The resulting statistical deviation had been caused by the loss of two samples from the 40°C group on day 21, and reduced the statistical precision of that treatment at the final time point, increasing the standard deviation range and potentially missing effects that may have been detectable with a more complete dataset. Beyond sample loss, MSP is binary and only detects whether methylation is present or absent, meaning subtle changes in epigenetic regulation across treatment groups may have gone undetected. Additionally, the observation window spanned only three weeks, which may have been too short to capture the full trend of growth, as the data did not show clear evidence of a plateau or decline in growth rate across either heat treatment.

Future studies should investigate the combined effects of salinity stress and temperature stress on DNA methylation patterns in Spirodela polyrhiza because our results suggest that thermal stress alone drove demethylation, raising the question of whether other environmental stressors associated with climate change produce similar or compounding effects. Particularly, salinity given its increasing prevalence in freshwater ecosystems due to agricultural runoff and sea-level intrusion. A potential experiment would be designing a two factor experiment in which salinity level and temperature are manipulated independently across treatment groups. A two-way ANOVA would separate the effect of each stressor and detect any interaction between them. If both salinity and heat independently drive demethylation, the results would build a more realistic model of how duckweed responds to the compounding stressors that climate change produces.

Understanding how aquatic plants respond to thermal stress at the molecular level is important because epigenetic changes like DNA methylation can be inherited across generations (Dainelli et al., 2024). For a species like Spirodela polyrhiza, this has real consequences; a thermally stressed population that inherits altered methylation patterns at essential growth-related genes may be less fit as temperatures continue rising, potentially driving population decline that ripples through the freshwater food webs it supports. These findings suggest that global warming poses a threat to duckweed not just through direct heat damage, but through a heritable suppression of the genes that regulate normal growth. This has ramifications for aquatic ecosystem management and bioremediation, where duckweed is actively used to filter excess nutrients from water. Ultimately, the results of this study reveal that the biological cost of climate change may already be written into the epigenomes of the organisms we may not know we rely on most.

References:

Dainelli, M., Castellani, M. B., Pignattelli, S., Falsini, S., Ristori, S., Papini, A., Colzi, I., Coppi, A., & Gonnelli, C. (2024). Growth, physiological parameters and DNA methylation in Spirodela polyrhiza (L.) Schleid exposed to PET micro-nanoplastic contaminated waters. Plant Physiology and Biochemistry, 207, 1.

Hoagland, D. R., & Arnon, D. I. (1950). The water-culture method for growing plants without soil. California Agricultural Experiment Station, Circular 347. University of California, Berkeley.

Kakoulidou, I., Avramidou, E. V., Baránek, M., Brunel-Muguet, S., Farrona, S., Johannes, F., Kaiserli, E., Lieberman-Lazarovich, M., Martinelli, F., Mladenov, V., Testillano, P. S., Vassileva, V., & Maury, S. (2021). Epigenetics for crop improvement in times of global change. Biology, 10 (8), 766. https://doi.org/10.3390/biology10080766

Appenroth, K.-J., & Nickel, G. (2010). Turion formation in Spirodela polyrhiza: The environmental signals that induce the developmental process in nature. Physiologia Plantarum, 138 (3), 312–320. https://doi.org/10.1111/j.1399-3054.2009.01319.x

Kuehdorf, K., & Appenroth, K.-J. (2011). Influence of salinity and high temperature on turion formation in the duckweed Spirodela polyrhiza. Aquatic Botany, 97 (1), 69–72. https://doi.org/10.1016/j.aquabot.2011.10.003

Lopez, M.-E., Denoyes, B., & Bucher, E. (2024). Epigenomic and transcriptomic persistence of heat stress memory in strawberry (Fragaria vesca). BMC Plant Biology, 24, 405. https://doi.org/10.1186/s12870-024-05093-6

Naydenov, M., Baev, V., Apostolova, E., Gospodinova, N., Sablok, G., Gozmanova, M., & Yahubyan, G. (2015). High-temperature effect on genes engaged in DNA methylation and affected by DNA methylation in Arabidopsis. _Plant Physiology and Biochemistry_, _87_, 102-108. https://doi.org/10.1016/j.plaphy.2014.12.022

Van Antro, M., Prelovsek, S., Ivanovic, S., Gawehns, F., Wagemaker, N. C. A. M., Mysara, M., Horemans, N., Vergeer, P., & Verhoeven, K. J. F. (2023). DNA methylation in clonal duckweed (Lemna minor L.) lineages reflects current and historical environmental exposures. Molecular Ecology, 32 (2), 428–443. https://doi.org/10.1111/mec.16757